

Advanced Natural Language Processing

Discussion 9: Distillation, Compression, Pruning

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Computational Complexity of Transformers (and all other models in the Transformer Zoo)

Layer Type	Complexity per Layer	Sequential Operations	Maximum Path Length
Self-Attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Convolutional	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k(n))$
Self-Attention (restricted)	$O(r \cdot n \cdot d)$	$O(1)$	$O(n/r)$

n = sequence length (40-70),
 d = depth (number layers),
 k = kernel size of the convolution,
 d = length word embedding (128-512)
 r = attention window size

Model	Hardware	Power (W)	Hours	kWh-PUE	CO ₂ e	Cloud compute cost
Transformer _{base}	P100x8	1415.78	12	27	26	\$41–\$140
Transformer _{big}	P100x8	1515.43	84	201	192	\$289–\$981
ELMo	P100x3	517.66	336	275	262	\$433–\$1472
BERT _{base}	V100x64	12,041.51	79	1507	1438	\$3751–\$12,571
BERT _{base}	TPUv2x16	—	96	—	—	\$2074–\$6912
NAS	P100x8	1515.43	274,120	656,347	626,155	\$942,973–\$3,201,722
NAS	TPUv2x1	—	32,623	—	—	\$44,055–\$146,848
GPT-2	TPUv3x32	—	168	—	—	\$12,902–\$43,008

Table 3: Estimated cost of training a model in terms of CO₂ emissions (lbs) and cloud compute cost (USD).⁷ Power and carbon footprint are omitted for TPUs due to lack of public information on power draw for this hardware.

Other approaches to make LLM more efficient

- TinyBERT: transfer the (plenty) of knowledge encoded in large-BERT to a small student tinyBERT.
- DistilBERT: DistilBERT is a small, fast, cheap and light Transformer model trained by distilling BERT base. It has 40% less parameters than *bert-base-uncased*, runs 60% faster while preserving over 95% of BERT's performances as measured on the GLUE language understanding benchmark
- Quantization: e.g.: dynamic quantization converts a floating-point model to a quantized model with static int8 or float16 data types for the weights and dynamic quantization for the activations. No need for 64 bit floats in your weights, 8 or 16 bit will do. Leads easily to 25-250% speed up (depends on hardware)
- Pruning: An alternative to quantization is pruning. Pruning introduces zeros (aka sparsity) in the weight matrices, leading to more efficient matrix calculations in terms of both memory and computation. One can either prune “weights” or entire “neurons”.
- ... this is an on-going field of research ...

NEWS | 05 October 2022

DeepMind AI invents faster algorithms to solve tough maths puzzles

Machine-learning technique improves computing efficiency and could have far-reaching applications.

KV caching , Flash-attention, etc



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Improving efficiency

- Algorithmic optimizations (KV Cache, etc.)
- Memory access optimization (Flash Attention)
- Quantization
- Low-Rank Adaptation (LoRA)
- Distilling into smaller models
- Better training and data
- New architectures



Questions?



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Any Questions on the Lecture

Now let's form groups of 4



Discuss

- Quality loss after these type of operations
- Use for preventing adversarial attacks
- https://www.wsj.com/tech/ai/is-china-stealing-american-ai-why-distillation-has-washington-up-in-arms-f924d29e?st=raGw7d&reflink=article_email_share
- https://www.theatlantic.com/international/archive/2025/02/deepseek-ai-china-tech/681553/?utm_campaign=atlantic-intelligence&utm_content=20250207&utm_source=newsletter&utm_medium=email&lctg=6050e37c1fc16d137f862465&utm_term=Atlantic%20Intelligence

Discuss Exercises Lecture

Exercise 1: Knowledge Distillation Loss and Soft Labels (DistilBERT)

Knowledge distillation transfers representations from a large teacher model to a compact student network by training the student on both hard ground-truth labels and smoothed "soft labels" generated by the teacher.

- In the lecture, we specify the total loss function for training DistilBERT:

$$\text{total loss} = \alpha \cdot \mathcal{L}_{\text{student}} + (1 - \alpha) \cdot \mathcal{L}_{\text{distillation}}$$

Identify what each loss term represents, and explain why Kullback-Leibler (KL) divergence is chosen for $\mathcal{L}_{\text{distillation}}$.

- The soft predictions and soft labels are generated using softmax with temperature $T > 1$:

$$q_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

Why does scaling logits by temperature $T > 1$ expose rich "dark knowledge" (e.g., secondary class probabilities) that standard one-hot hard labels completely erase?

- According to Sanh et al. (2019), what percentage of parameters is removed and how much faster is DistilBERT compared to Bert-base-uncased, and how much of BERT's original GLUE performance is retained?



Exercise 2: Parameter-Efficient Fine-Tuning via LoRA and Quantization

Low-Rank Adaptation (LoRA) and quantization techniques allow adaptation and compression of multi-billion parameter models under tight hardware constraints.

- In LoRA (slides 26 to 28), an original frozen projection matrix $W_Q \in \mathbb{R}^{M \times N}$ is updated using low-rank decomposition matrices A_q and B_q :
$$W_Q \leftarrow W_Q + A_q B_q^T$$
 - If $M = 4096$, $N = 4096$, and rank $r = 8$, calculate the number of trainable parameters in $A_q B_q^T$ versus W_Q . What is the parameter reduction ratio?
 - How are A_q and B_q initialized at the beginning of training to ensure the model's pre-trained behavior is initially unaltered?
- Contrast **Post-Training Quantization (PTQ)** with **Quantization-Aware Training (QAT)**. Why does reducing weights from 32-bit/16-bit floating point to 8-bit integers (or 1.58-bit ternary values in BitNet b1.58) lead to significant energy savings and hardware speedups?
- Explain the difference between weight pruning and neuron/head pruning. How does introducing structured sparsity assist in edge deployment for resource-constrained environments (e.g., wearables or medical implants)?

Exercise 3: Hardware-Aware Attention Optimization (KV Caching & FlashAttention)

During autoregressive decoding, token-by-token generation faces major memory bandwidth and compute bottlenecks.

- In the lecture, we demonstrate that during autoregressive generation of token x_{k+1} , previously generated tokens $x_0, \dots, x_k$ remain fixed.
 - Explain what **KV Caching** stores across decoding steps.
 - Why does caching Key and Value projections eliminate redundant matrix multiplications, and what memory challenge does it introduce inside GPU memory as context length grows?
- Using the GPU memory hierarchy shown on slides 12 to 14 (SRAM vs. HBM):
 - Contrast the memory size and bandwidth of on-chip SRAM versus High Bandwidth Memory (HBM) on an NVIDIA A100.
 - In standard attention, why do intermediate reads and writes of the $N \times N$ attention matrix ($S = QK^T$ and $P = \text{softmax}(S)$) to HBM create an I/O bottleneck?
- How does **FlashAttention** overcome this memory traffic bottleneck using kernel fusion and matrix tiling? Why does computing exact softmax over tiled blocks entirely within fast SRAM without materializing the full attention matrix in HBM achieve such drastic wall-clock speedups?

Discussion



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On Distillation: Is Knowledge Something That Can Be Handed Down?

A student with half the layers reproduces the teacher's decisions almost perfectly, by learning not the right answers but the teacher's entire distribution: what it found plausible, what it confused, how sure it was.

If the knowledge fits in half the parameters, was the teacher twice too big? Or does the student only inherit the teacher's habits, including its errors and doubts, without the capacity that was needed to form them?

A teacher can only be distilled once it exists. If every compact model is a copy of a large one, is compression a way to avoid training giants, or a way to make sure we always need them? And what is lost when a model learns from a model instead of from the world?

On Efficiency: Who Pays for Intelligence, and Does Cheaper Mean Less?

Training one large model consumes as much energy as a small town for weeks (slide 2). Every technique in this lecture makes that cheaper: distillation, quantisation, caching, better kernels. Cheaper has never once meant less: each saving has so far been spent on bigger models and more use.

Is efficiency an engineering virtue, an environmental obligation, or just the fuel of the next scale-up? (Jevons noticed in 1865 that more efficient steam engines increased the consumption of coal.)

The cost of a model is paid now, by everyone, through energy and water; the benefit is collected later, by some. Who gets to decide that the trade is worth it, and is “we made it 60 percent cheaper” an answer to that question or a way of not hearing it?

On the Return of Recurrence: Does NLP Progress, or Go in Circles?

This course told a story: recurrence was replaced by attention because recurrence was a bottleneck and could not be parallelised. Now Mamba and xLSTM train in parallel, scale linearly, and match Transformers on many tasks. Recurrence is back.

Was the move to attention a discovery or a detour? Does progress in this field accumulate, or circle, with each generation rediscovering what the previous one abandoned, under new names and with new hardware?

If “obsolete” turned out to mean “inconvenient on the chips of 2014”, how much of what you were taught as settled in this course should you expect to see return, and how would you tell a real advance from a fashion?

Tutorials Distillation

Compression Pruning



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Questions?



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